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An Air Accident Investigator's Perspective

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Abstract

While most investigations find evidence suggesting why mistakes happened, in some cases it seems impossible to envisage why flight crew made the mistakes that they did. Two accidents are briefly described. In each of the cases it is difficult to envisage how the experienced crew could have made the catastrophic errors that they did, unless they were impaired in some way. It appears possible that this could have been the result of ingesting oil products.

Keywords

cabin air, fumes, contamination, oil, mental capacity, error, aircraft, crash, accident

1 Introduction

Multiple cases with piston-engine aircraft where carbon monoxide poisoning caused pilot impairment and/or incapacitation have occurred. Additionally, there is an appreciable incidence of airliners suffering contamination of cabin air with fumes, often apparently due to oil entering the air bled from engines or APU (Auxiliary Power Unit) for cabin pressurisation and conditioning. It appears that synthetic oils, containing organophosphate compounds, are widely used in engines and APUs.

In some of the fume events flight crews have reported experiencing severe degradation in their mental capacity and performance, and only becoming aware of this after starting to use oxygen masks.

It appears that toxicological assessments at post mortem during accident investigations do not often include looking for signs of oil product ingestion. This raises the question as to whether cabin air contamination with oil products might insidiously have seriously impaired the crew, and consequently have been a significant causal factor in some apparently inexplicable accidents. Perhaps accident investigators have generally paid insufficient attention to this possibility?

2 Boeing 727, Flight DA1008, 25 April 1980, Tenerife, Canary Islands, Spain

A possible example: Boeing 727, Flight DA1008 – crash during a landing attempt at Tenerife Norte Airport in 1980. (Spanish Civil Aviation Accident Commission 1981).

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The flight crew consisted of two pilots and a Flight Engineer. The Commander had an appreciable amount of flight experience. He had flown to Tenerife Norte 58 times, and the Co-Pilot 9 times. It is inescapable that the crew knew that there was serious high ground around the airport, with the 12,188 ft Mt Teide only a relatively short distance away. DA1008 descended towards the airport, navigating using radio beacons, probably mostly in cloud, intending to conduct a procedural approach based on an NDB (Non-Directional Beacon, a type of radio beacon). Deviations in air-speed, flight path, and position reporting, suggested crew performance may have been degraded.

The Air Traffic Controller (ATC) made a number of serious errors, including clearing DA1008 to descend to a dangerously low altitude, and unexpectedly instructing it to enter a holding pattern based on an NDB, using contradictory and confusing terms. The hold was unpublished, and therefore not on the crew's charts.

None of the crew members knew how to enter the hold. They discussed the matter for over 2 minutes, plainly showing they knew that they did not know, while wandering round to high ground. DA1008 was at 6,000 ft altitude, while in a sector with a minimum safe altitude of 14,500 ft.

The Ground Proximity Warning System (GPWS) activated. The crew applied full thrust, but did not level the wings, as specified by the procedure for a GPWS warning, and this seriously degraded the climb performance. The aircraft impacted the mountains, killing all on board.

3 Boeing 757, Flight AA965, 20 December 1995, Valle del Cauca, Colombia

A further possible example: A Boeing 757, Flight AA965, which crashed during a night landing approach to Cali, Colombia in 1995. (Aeronautica Civil of The Republic of Colombia 1996, Ladkin 1996).

The pilots were qualified and highly experienced and must have known of the mountainous terrain around Cali. Their inadvertent selection of an incorrect radio beacon caused a major departure of the aircraft from the intended flight path along a valley leading to the airport. On realising this, while knowing that they were uncertain about their position, the crew headed directly back towards the intended flight path. They did not seek assistance from ATC.

On approaching high ground, a GPWS warning triggered. The crew executed the escape manoeuvre, but omitted to retract airbrakes, severely degrading climb performance. The aircraft impacted the mountains, killing a number of the occupants.

4 Summary

In each of the above cases it is difficult to envisage how the experienced crew could have made the catastrophic errors that they did, unless they were impaired in some way. It appears possible that this could have been the result of ingesting oil products.

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About the Author

Tony Cable received his degree in aeronautical engineering from the University of London, United Kingdom.

He worked for Boeing and British Aerospace for a total of 9 years. As an Aircraft Accident Investigation Consultant, he served 32 years with the Air Accidents Investigation Branch (AAIB), investigating the engineering aspects of accidents and incidents to many types of fixed wing and rotary-wing aircraft with the aim of improving flight safety. These

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